

Jacob Ortiz

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EDUCATION

University of California San Diego | La Jolla, CA

B.S. in Physics

June 2026

Relevant Courses: Math & Comp Physics, Quantum Physics, StatMech, Data Science in Practice, IBM: Gen AI Applications w/ RAG & LangChain, Fundamentals of AI Agents Using RAG & LangChain, Gen AI Engineering & Fine-Tuning Transformers, Gen AI Advanced Fine-Tuning for LLMs, Gen AI Language Modeling with Transformers, Gen AI: Intro & Applications, Gen AI: Prompt Engineering Basics, Gen AI & LLMs: Architecture & Data Preparation, Gen AI Foundational Models for NLP & Language Understanding, Intro to Deep Learning & Neural Networks w/ Keras, Machine Learning w/ Python, Developing AI Apps w/ Python & Flask, Building Gen AI-Powered Apps w/ Python, Data Analysis, Python for Data Science, AI, & Development

TECHNICAL SKILLS

Languages and Tools: Python, Matlab, Mathematica, R Studio, LaTeX, Markdown, HTML, C, CSS, JavaScript, SQL

Developer Tools: Git, GitHub, Kubernetes, Docker, Nautilus, Arduino, Zapier, Microsoft Copilot Studio, OpenAI GPT

Frameworks and Libraries: Pytorch, TensorFlow, Keras, NRP Nautilus, HuggingFace, LangChain, XGBoost, spaCy, NLTK, TorchText, Scikit-Learn, Statsmodels, SciPy, NumPy, Jupyter, Pandas, Pygame, Matplotlib, Seaborn, Plotly, ChromaDB, Watsonx, Gradio, Flask

EXPERIENCE

AI Researcher | AMERICAN REFRIGERATION | REMOTE

August 2025 - Present

- Led analytical and technical evaluation of AR automation platforms, produced executive recommendations adopted by leadership, quantifying 82% step reduction, 48% productivity gain, 12–30% AR aging reduction, ~\$1M projected cash-flow improvement.
- Built and deployed 9 AI automation systems w/ **Python, Copilot Studio, GPT Team, Zapier** enabling cross-department workflow.
- Automated weekly analysis, inbox triage, meeting intelligence, onboarding workflows to save 13–20 hours/week (~\$35k–\$65k/year).
- Delivered executive-ready summaries, automated task routing, and structured data logs.

ML Research Assistant | UCSD DUARTE LAB | HYBRID

February 2025 - Present

- Co-authored NeurIPS 2025 ML4PS paper & poster “Why is Attention Sparse in Particle Transformer?” (arXiv:2512.00210v1).
- Built transformer based models in **PyTorch & TensorFlow**, preprocessing & inference pipelines, GPU accelerated experiments.
- Improved efficiency and interpretability, maintaining performance with novel attention mechanisms and model architectures.
- Performed interpretability studies analyzing pre softmax attention, substructure signals, and physics inspired inductive biases.

PERSONAL PROJECTS

Custom Lightweight CNN Medical Imaging

December 2025

- Implemented efficient CNN inspired by Eff-PCNet architecture, benchmarked against EfficientNetV2-S on medical datasets
Engineered full training pipeline (EDA, preprocessing, augmentation, architecture, eval) in **TensorFlow/Keras**
- Reduced parameter count 5x (20.3M to 4.39M) while improving performance (+20.4% accuracy, +21.8% AUC on SkinCancer; +9.1% accuracy, +36.4% AUC on HAM10000).
- Designed and implemented custom M2C and Rep-C blocks, achieving competitive accuracy under strict memory limits.

AI RAG Assistant

November 2025

- Built a full RAG pipeline with **LangChain, ChromaDB**, and **Watsonx** Mixtral-8x7B to automate scientific literature analysis.
- Implemented document loaders, recursive text splitting, embeddings, and a custom retriever, improving query accuracy by ~65%.
- Developed a real-time PDF QA bot enabling researchers to extract insights instantly, reducing manual review time by 70%+.

Transformer Fine Tuning

October 2025

- Fine-tuned transformer models using **HuggingFace**, LoRA, and PEFT for domain-shifted text classification (AG News to IMDB).
- Built efficient, low-parameter training pipelines with preprocessing, tokenization, custom training loops, eval, and inference.
- Improved sentiment classification generalization, reducing GPU memory, skill in LLM fine-tuning and scalable PEFT workflows.

Synthesizer Transformer Implementation and Benchmarking on Nautilus

July 2025

- Implemented Dense Synthesizer attention (arXiv:2005.00743) in **TensorFlow**, replacing dot-prod attention with learned matrices.
- Benchmarked models on the hls4ml jet-class. dataset, evaluating accuracy, FLOPs, memory use, speed, and ROC AUC on clusters.
- Achieved up to 18% faster inference with competitive accuracy, contributing to research on ML for particle physics.